

Spring & Spring Boot Viva Questions

Accenture Full Stack + GenAI Viva preparation — Spring backend only. Covers Spring Core, Spring MVC, Spring Boot, Spring Data JPA, REST, validation, AOP and testing. Excludes Docker, Agile, Kubernetes, frontend and other non-Spring topics.

1. Spring Core

1. What is Spring?
2. Why do we use Spring?
3. What are the major features of Spring?
4. What is IoC?
5. What is Dependency Injection?
6. IoC vs Dependency Injection?
7. What is a Spring Container?
8. What is a Spring Bean?
9. How is a Bean created?
10. What is ApplicationContext?
11. ApplicationContext vs BeanFactory?
12. What is component scanning?
13. What is autowiring?
14. What are the types of dependency injection?
15. Constructor injection vs setter injection?
16. Why is constructor injection preferred?
17. What is @Autowired?
18. What is @Qualifier?
19. What is @Primary?
20. What is @Component?
21. What is @Service?
22. What is @Repository?
23. @Component vs @Service vs @Repository?
24. What is @Configuration?
25. What is @Bean?
26. @Bean vs @Component?
27. What are Spring Bean scopes?
28. Singleton vs Prototype?
29. What is Bean lifecycle?
30. What is @PostConstruct?
31. What is @PreDestroy?

2. Spring MVC

- 32. What is Spring MVC?
- 33. What is MVC architecture?
- 34. What is DispatcherServlet?
- 35. What is a Controller?
- 36. What is @Controller?
- 37. What is @RestController?
- 38. @Controller vs @RestController?
- 39. What is @RequestMapping?
- 40. @GetMapping vs @PostMapping?
- 41. What is @PutMapping?
- 42. What is @DeleteMapping?
- 43. What is @PathVariable?
- 44. What is @RequestParam?
- 45. What is @RequestBody?
- 46. What is @ResponseBody?
- 47. What is ResponseEntity?
- 48. How does a request flow through Spring MVC?

3. Spring Boot

- 49. What is Spring Boot?
- 50. Spring vs Spring Boot?
- 51. Why is Spring Boot preferred?
- 52. What is auto-configuration?
- 53. What is @SpringBootApplication?
- 54. What are Spring Boot starter dependencies?
- 55. What is an embedded server?
- 56. Which embedded server does Spring Boot use by default?
- 57. What is application.properties?
- 58. What is application.yml?
- 59. How do you configure a database in Spring Boot?
- 60. What are Spring Profiles?
- 61. What is @Profile?
- 62. What is externalized configuration?
- 63. What is Spring Boot Actuator?
- 64. What are Spring Boot runners?
- 65. CommandLineRunner vs ApplicationRunner?
- 66. How does Spring Boot start an application?

4. Spring Data JPA

- 67. What is Spring Data JPA?
- 68. Why do we use Spring Data JPA?
- 69. What is JpaRepository?
- 70. CrudRepository vs JpaRepository?
- 71. What is a Repository?
- 72. How does Spring Data create repository implementations?
- 73. What are derived query methods?
- 74. What is @Query?
- 75. What is JPQL?
- 76. JPQL vs native SQL?
- 77. What is pagination?
- 78. What is sorting?
- 79. How do you update/delete using Spring Data JPA?

5. REST API

- 80. What is REST?
- 81. What is a REST API?
- 82. What are REST principles?
- 83. GET vs POST?
- 84. PUT vs PATCH?
- 85. DELETE?
- 86. What are HTTP status codes?
- 87. 200 vs 201?
- 88. 400 vs 401 vs 403 vs 404?
- 89. What is 500?
- 90. What is idempotency?
- 91. What is JSON?
- 92. How does Spring Boot convert Java objects to JSON?
- 93. What is Jackson?
- 94. How do you handle exceptions globally?
- 95. What is @ControllerAdvice?
- 96. What is @ExceptionHandler?

6. Validation

- 97. What is validation?
- 98. What is Bean Validation?
- 99. What is @Valid?

- 100.** What is @NotNull?
- 101.** @NotBlank vs @NotEmpty?
- 102.** What is @Size?
- 103.** How do you handle validation errors?
- 104.** Why should validation be performed on API input?

7. Spring AOP

- 105.** What is AOP?
- 106.** Why do we use AOP?
- 107.** What is a cross-cutting concern?
- 108.** What is an Aspect?
- 109.** What is a Join Point?
- 110.** What is Pointcut?
- 111.** What is Advice?
- 112.** Types of Advice?
- 113.** What is @Before?
- 114.** What is @After?
- 115.** What is @Around?

8. Spring Testing

- 116.** What is unit testing?
- 117.** What is integration testing?
- 118.** What is JUnit?
- 119.** What is Mockito?
- 120.** @Mock vs @MockBean?
- 121.** What is @SpringBootTest?
- 122.** What is @WebMvcTest?
- 123.** Why do we use mocks?